TUT-4 19/09/20

① ⓐ As the motor stopped, the only horizontal force acting on boat is — $f_D = -bv$

$$\Rightarrow m\frac{dv}{dt} = -bv$$

$$\Rightarrow \int \frac{dv}{v} = \int -\frac{b}{m}dt$$

$$\Rightarrow \ln v = -\frac{b}{m}t + \ln v_0$$

$$\Rightarrow \ln(v/v_0) = -b/m\, t \Rightarrow \boxed{v = v_0 e^{-\frac{bt}{m}}}$$

$$\Rightarrow \boxed{v = v_0 e^{-b/m\, t}}$$

Now

$$\frac{ds'}{dt'} = v_0 e^{-\frac{b}{m}t}$$

$$\Rightarrow \int_0^x ds' = \int_0^t v_0 e^{-b/m\, t'} dt'$$

$$\cancel{\Rightarrow s = v_0} \Rightarrow \boxed{x = \frac{mv_0}{b}\left(1 - e^{-\frac{b}{m}t}\right)}$$

at $t \to \infty \Rightarrow e^{-b/m\, t} \to 0$

So, $\boxed{x_{max} = \frac{mv_0}{b}}$ ← maximum limit of displacement

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(b) Now $v_0 = 4\,m/s$

$v = 1\,m/s$

$t = 10\,s$

$$v = v_0 e^{-\frac{b}{m}t}$$

$$\Rightarrow 1 = 4 \cdot e^{-\frac{b}{m}10}$$

$$\Rightarrow -\ln(4) = -\frac{b}{m}(10)$$

$$\Rightarrow \frac{b}{m} = \frac{1}{10}\ln(4)$$

$$\Rightarrow \boxed{\frac{b}{m} = 0.14\,s^{-1}}$$

So, $x_{max} = \frac{m v_0}{b} = \frac{4\ m/s}{0.14\ s^{-1}} = 29\,m$

$\Rightarrow \boxed{x_{max} = 29\,m} \rightarrow$ limiting position.

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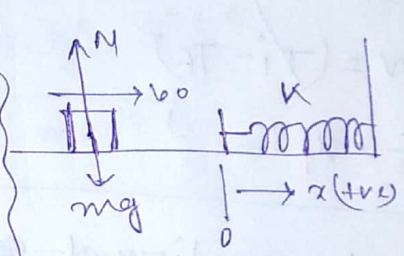
(3) we can slove it by work-energy theorem →



[Final energy = initial energy + work done by the non-conservative force. (frictional force here)] — (1)

Now, work done by frictional force -

$$W = \int_0^l f\,dx = \int_0^l \mu N\,dx$$

$$= \int_0^l \mu \cdot mg\,dx = \mu m g l.$$

Now initial energy $= K_i + U_i$

$$= \frac{1}{2} m v_0^2 + 0$$

Final energy $= K_f + U_f = 0 + \frac{1}{2} k l^2$ [$\because v_f = 0$]

(1) $\rightarrow \frac{1}{2} k l^2 = \frac{1}{2} m v_0^2 + \mu m g l$

$$\Rightarrow \boxed{l = v_0 \sqrt{\frac{m}{k + mgb}}}$$

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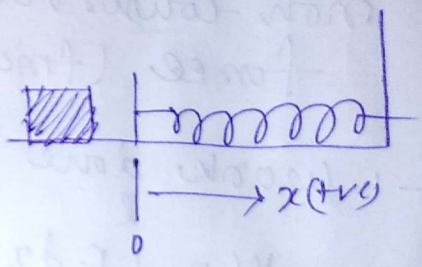
(2) Let, the position of ~~spm~~ spring when block touches is $x = 0$ and final position of block = x_f.



So, The work done by block against the spring —

$$W = \int_{r_i}^{r_f} \vec{F} \cdot d\vec{r} = \int_0^{x_f} (-kx\hat{i}) \cdot (dx\hat{i})$$

$$= \int_0^{x_f} (-kx)\,dx$$

$$\Rightarrow \boxed{W = -\tfrac{1}{2} k x_f^2}$$

Now work done = change in K.E = $K_f - K_i$

$$\Rightarrow -\tfrac{1}{2} k x_f^2 = 0 - \tfrac{1}{2} m v_i^2$$

$$\Rightarrow \boxed{x_f = v_i \sqrt{m/k}} \Rightarrow \boxed{x_f = 7.4 \text{ cm}}$$ → putting the values.

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(4) $v = k\sqrt{s}$

$$\Rightarrow \frac{dv}{dt} = \frac{k}{2\sqrt{s}}\left(\frac{ds}{dt}\right) = \frac{k}{2\sqrt{s}}\, v$$

$$= \frac{k}{2\cancel{\sqrt{s}}}\, k\cancel{\sqrt{s}}$$

$$= \frac{k^2}{2}$$

$$\Rightarrow v = \frac{k^2}{2}t + \underset{\Downarrow\, 0}{c_1} \qquad [t=0,\ v=0 \Rightarrow c_1 = 0]$$

$$\Rightarrow \boxed{v = \frac{k^2}{2}t}$$

$$\Rightarrow \frac{ds}{dt} = \frac{k^2}{2}t$$

$$\Rightarrow s = \frac{k^2 t^2}{4} + \underset{\Downarrow\, 0}{c_2} \qquad [t=0,\ s=0 \Rightarrow c_2 = 0]$$

$$\Rightarrow \boxed{s = \frac{k^2 t^2}{4}}$$

Now, work done $W = F \cdot s$

$$= ma \cdot s$$

$$= m\frac{dv}{dt} \cdot s$$

$$= m \cdot \frac{k^2}{2} \cdot \frac{k^2 t^2}{4}$$

$$\boxed{W = \frac{1}{8} m k^4 t^2}$$

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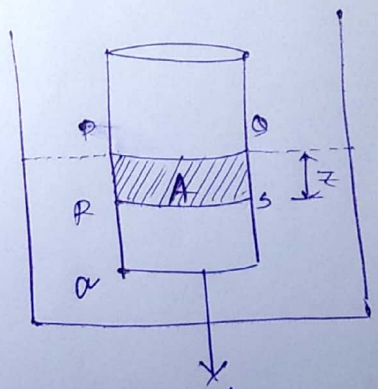
5. Let, R_s, the equilibrium position of the cylinder.

Let, it is pushed to a height z.



By, Archimedes principle the buoyant force acting on cylinder is = weight of displaced fluid

$$= \sigma(A \cdot z)\, g$$

$$mg = W$$

$$\Rightarrow m = \frac{W}{g}$$

Now by Newton's 2nd Law,

$$m\frac{dv}{dt} = \sigma(A \cdot z)\, g$$

$$\Rightarrow \frac{W}{g}\frac{dv}{dt} = \sigma(Az)\, g \Rightarrow \boxed{\frac{d^2 z}{dt^2} + \frac{g^2 A \sigma}{W} z = 0}$$

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$$\Rightarrow \frac{d^2z}{dt^2} + \omega_1^2 z = 0 \quad ; \quad \omega_1 = \sqrt{\frac{g^2 A \sigma}{W}}$$

$$\Rightarrow (D^2 + \omega_1^2) z = 0 \qquad D^2 = \frac{d^2}{dt^2}$$

<u>C.F</u>, $m^2 + \omega^2 = 0 \qquad D \equiv m$

$$\Rightarrow \boxed{m = \pm i\omega_1}$$

$$\therefore z(t) = c_1 \cos \omega_1 t + c_2 \sin \omega_1 t$$

$$= c_1 \cos \sqrt{\frac{g^2 A \sigma}{W}}\, t + c_2 \sin \sqrt{\frac{g^2 A \sigma}{W}}\, t$$

<u>So</u>, time period of oscillation,

$$\boxed{T = \frac{2\pi}{\omega_1} = 2\pi \sqrt{\frac{W}{g^2 A \sigma}}}$$

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